Investigation of non-Newtonian fluids under oscillatory forcing

Alexander Contreras, Dr. Joseph Kalman

Department of Mechanical and Aerospace Engineering

Solid Propulsion and Combustion Laboratory (SPACL)

Hung Family College of Engineering, Long Beach State

Introduction:

Solid rocket propellants are mixed from a viscous polymer binder (HTPB) and a heavy load of solid oxidizer (ammonium perchlorate). How well they mix sets burn-rate uniformity, void content, and motor safety [1].

Resonant Acoustic Mixing (RAM) does this without a blade — it shakes the whole vessel vertically near a ~60 Hz resonance, mixing through acoustically driven interfacial motion instead of an impeller [2].

In 2022, Huo & Wang modeled how RAM deforms a non-Newtonian fluid under low-frequency vertical forcing [2].

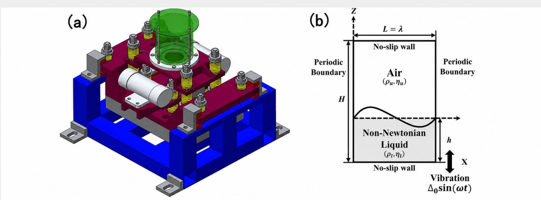

Figure 1: *Resonant Acoustic Mixing of a non-Newtonian fluid (a) RAM apparatus and (b) simplified physical model (gray = power-law non-Newtonian fluid, white = air) under vertical oscillation.*

RAM mixes by folding and renewing the interface; how easily it does this depends on the fluid's rheology [2].

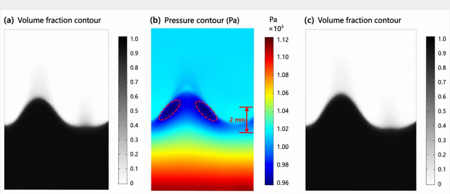

Figure 2: *Simulated Faraday-wave interface — the folding that drives RAM mixing.*

Binder rheology is set by composition, but its effect on droplet spreading — the mixing proxy — is uncharacterized.

Objective:

To measure how an HTPB droplet spreads under resonant acoustic forcing as its rheology is varied by n-dodecane dilution, revealing the role of binder rheological properties in mixing.

Experimental & Analysis Method:

(1) Samples. HTPB (LBHP) diluted with n-dodecane to 10%, 18.5%, 30%, 38.5%. Dilution spans a range of binder rheologies.

(2) Imaging. Droplet on the exciter at fixed forcing, side profile at 61,200 fps. The reflective surface mirrors the droplet below the contact line.

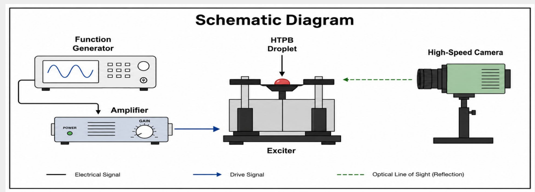

Figure 3: *Function Generator → Amplifier → Exciter (droplet) → High-Speed Camera.*

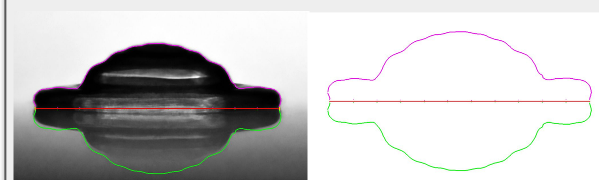

Figure 4: *Figure 4: raw frame (left) and processed outline (right).*

(3) Tracking (MATLAB). Contact line found as the mirror-symmetry axis; droplet, reflection, and contact line outlined [3]. Extends dynamic contact-angle work on the LabRAM from our group.

Data Reduction — Quantifying Spread:

Free-fall sets the scales D_0 (diameter) and U_0 (velocity). Surface spreading gives D(t), made unitless and fit with a two-stage power law:

$$S^* = \frac{D(t)}{D_0} \qquad t^* = \frac{U_0}{D_0}\,t$$

Acknowledgements:

This project was funded by the National Science Foundation (LSAMP, NSF Grant #1826490) and the Chancellor's Office of the California State University. Thank you to Dr. Joseph Kalman and the Solid Propulsion and Combustion Laboratory (SPACL) at CSULB.

Results:

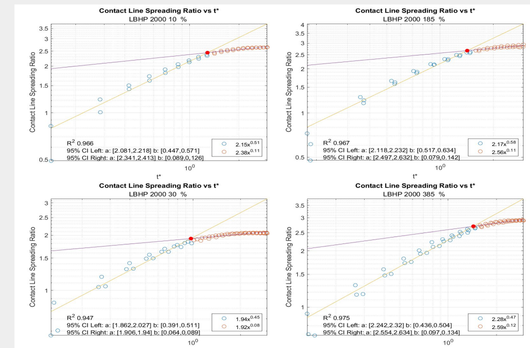

Figure 5: *Spreading ratio S* vs. dimensionless time t* for four dilutions, with two-stage power-law fits (R^2 = 0.95–0.98).*

Perspectives:

Constant exponents → the spreading mechanism is the same regardless of rheology (fast inertio-capillary, then slow viscous/equilibrium).

Composition controls how far the droplet spreads — i.e., how much interface is created. The non-monotonic trend points to a composition-dependent mixing optimum: 30% spreads least, leaner and richer mixtures spread more.

Caveat: four samples at one forcing condition, and S is an interfacial proxy, not bulk mixing. Future work: more concentrations and varied forcing.*

References:

[1] McDonald, B. A., Rice, J. R., & Kirkham, M. W. (2014). Humidity induced burning rate degradation of an iron oxide catalyzed ammonium perchlorate/HTPB composite propellant. Combustion and Flame, 161(1), 363–369. https://doi.org/10.1016/j.combustflame.2013.08.014

[2] Huo, Q., & Wang, X. (2022). Faraday instability of non-Newtonian fluids under low-frequency vertical harmonic vibration. Physics of Fluids, 34(9). https://doi.org/10.1063/5.0108295

[3] Marquez, I., & Kalman, J. (2023). Dynamic Contact Angle Measurements Using LabRAM. AIAA SCITECH 2023 Forum. https://doi.org/10.2514/6.2023-2558